

Cartan Criteria

1.1 Jordan Decomposition

Lemma 1.1.1

Let x be a linear transformation of the complex vector space V . Suppose that x has Jordan decomposition $x = d + n$, where d is diagonalisable, n is nilpotent, and d and n commute.

- (a) There is a polynomial $p(X) \in \mathbb{C}[X]$ such that $p(x) = d$.
- (b) Fix a basis of V in which d is diagonal. Let $\bar{d}$ be the linear map whose matrix with respect to this basis is the complex conjugate of the matrix of d . There is a polynomial $q(X) \in \mathbb{C}[X]$ such that $q(x) = \bar{d}$.

Exercise 1.1.2

Let V be a vector space, and suppose that $x \in \mathfrak{gl}(V)$ has Jordan decomposition $d + n$. Show that $\text{ad } x : \mathfrak{gl}(V) \rightarrow \mathfrak{gl}(V)$ has Jordan decomposition $\text{ad } d + \text{ad } n$.

Proof. Only need to show that $\text{ad } d$ is semisimple and $\text{ad } n$ is nilpotent, then the proposition follows from the linearity of ad and the uniqueness of Jordan decomposition.

- Suppose that $e_1, \dots, e_n$ is the basis in which d is represented diagonal, say $\text{diag}(\lambda_1, \dots, \lambda_n)$. Let $E_{ij} \in \mathfrak{gl}(V)$ be the maps that mapping e_j to e_i , that is $E_{ij}(e_k) = \delta_j^k e_i$, $k = 1, \dots, n$. Then

$$\begin{aligned}
 \text{ad}(d)(E_{ij})(e_k) &= [d, E_{ij}](e_k) \\
 &= (dE_{ij} - E_{ij}d)(e_k) \\
 &= d(E_{ij}e_k) - E_{ij}(de_k) \\
 &= d(\delta_j^k e_i) - E_{ij}(\lambda_k e_k) \\
 &= \delta_j^k \lambda_i e_i - \lambda_k \delta_j^k e_i \\
 &= (\lambda_i - \lambda_k) \delta_j^k e_i = (\lambda_i - \lambda_j) E_{ij}
 \end{aligned}$$

Which implies that E_{ij} is an eigenvector for $\text{ad}(d)$ for each $i, j = 1, \dots, n$.
 $\text{ad } n$ is diagonal under the basis $E_{ij}, i, j = 1, \dots, n$

- Since n is nilpotent, it follows that $\text{ad } n$ is nilpotent.
- It follows from Jacobi identity that

$$(\text{ad } d) \circ (\text{ad } n)(y) = [d, [n, y]] = -[y, [d, n]] - [n, [y, d]]$$

Since d, n commutes, $[d, n] = 0$. Thus $\text{ad } n$ and $\text{ad } d$ commute.

□

1.2 Testing for Solvability

Exercise 1.2.1

Let V be a complex vector space, L be a Lie subalgebra of $\mathfrak{gl}(V)$. Suppose that L is solvable. Use Lie's Theorem to show that there is a basis of V in which every element of L' is represented by a strictly upper triangular matrix. Conclude that $\text{tr } xy = 0$ for all $x \in L$ and $y \in L'$.

Remark.

$\mathbb{C}$ can be replaced by $\mathbb{F}$ such that $\text{char } \mathbb{F} = 0, \bar{\mathbb{F}} = \mathbb{F}$.

Proof. From Lie's Theorem, there is a basis of V in which every element of L is upper triangular, say $\{e_1, \dots, e_n\}$. Take $z \in L', z = [x, y], x, y \in L$. Then

$$z(e_1) = [x, y](e_1) = x(y(e_1)) - y(x(e_1)) = 0$$

Since e_1 is simultaneously an eigenvector of x and y . Let $L_k = \text{span}\{e_1, \dots, e_k\}$, then

$$z(e_k) = x(y(e_k)) - y(x(e_k))$$

Suppose,

$$y(e_k) = \sum_{j=1}^k y_{ij} e_j, \quad x(e_k) = \sum_{j=1}^k x_{ij} e_j$$

then

$$\begin{aligned} x(y(e_k)) + \text{span}\{e_1, \dots, e_{k-1}\} &= x(y_i e_k) + \text{span}\{e_1, \dots, e_{k-1}\} \\ &= x_i y_i e_k + \text{span}\{e_1, \dots, e_{k-1}\} \end{aligned}$$

Similarly,

$$y(x(e_k)) + \text{span}\{e_1, \dots, e_{k-1}\} = x_i y_i e_k + \text{span}\{e_1, \dots, e_{k-1}\}$$

Thus

$$z(e_k) \in \text{span}\{e_1, \dots, e_{k-1}\}$$

It follows that z is nilpotent. Thus every element of L' is represented by a strictly upper triangular matrix.

The matrix of x and y under the basis $\{e_1, \dots, e_n\}$ are upper triangular matrix and strictly upper triangular matrix, respectively, say A, B . Then

$$\text{tr}xy = \text{tr}(AB) = \sum_{k=1}^n a_{ik}b_{kj} = 0$$

Since for each $i > k$, $a_{ik} = 0$, and for each $i \leq k$, $b_{kj} \leq 0$. □

Proposition 1.2.2

Let V be a complex vector space and let L be a Lie subalgebra of $\mathfrak{gl}(V)$. If $\text{tr}xy = 0$ for all $x, y \in L$, then L is solvable.

Proof. We shall show that every element of L' is nilpotent, then it follows from Engel's Theorem that L' is nilpotent, thus L is solvable.

Take any element $x \in L'$. Suppose that x has a Jordan decomposition $x = d + n$. We choose a basis in which d is diagonal $d = \text{diag}(\lambda_1, \dots, \lambda_n)$ and n is strictly upper triangular. We only need to show that

$$\begin{aligned} \sum_{i=1}^n \lambda_i \bar{\lambda}_i &= 0 \\ \text{tr}(\bar{d}x) &= \text{tr}(\bar{d}d + \bar{d}n) = \text{tr}(\bar{d}d + \bar{d}n) \\ &= \text{tr}(\bar{d}d) + 0 \\ &= \sum_{i=1}^n \lambda_i \bar{\lambda}_i \end{aligned}$$

Since $x \in L'$, we write $x = [y, z]$, $y, z \in L'$.

$$\begin{aligned}\operatorname{tr}(\bar{d}x) &= \operatorname{tr}(\bar{d}[y, z]) \\ &= \operatorname{tr}(\bar{d}yz - \bar{d}zy) = \operatorname{tr}(\bar{d}yz - y\bar{d}z) \\ &= \operatorname{tr}([\bar{d}, y]z)\end{aligned}$$

Note.

这里我们折腾 $\operatorname{ad} \bar{d}$ 而非 $\bar{d}$ 的意义在于, 多项式对不变性的保持需要我们对结合代数封闭, 而李代数不一定保持结合代数的封闭性.

Thus to show $d = 0$, we only need to show that $[\bar{d}, y] \in L$, that is $\operatorname{ad}_{\bar{d}}(y) \in L$. $\operatorname{ad} x$ has Jordan decomposition, $\operatorname{ad} x = \operatorname{ad} d + \operatorname{ad} n$. Then from Lemma 1.1.1, there exists $q(X) \in \mathbb{C}[X]$ such that

$$q(\operatorname{ad} x) = \operatorname{ad} \bar{d}$$

Since L is $\operatorname{ad} x$ -invariant, so dose $q(\operatorname{ad} x)$ and $\operatorname{ad} \bar{d}$, which completes the proof. $\square$

Theorem 1.2.3

Let L be a complex Lie algebra. Then L is solvable if and only if $\operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) = 0$ for all $x \in L$ and all $y \in L'$.

Note.

由于 L 的 solvability 等价于任意阶导出子代数 $L^{(k)}$ 的 solvability, 因此命题中的

$$x \in L, \quad y \in L'$$

对于任意固定的 $k \geq 1, l \geq 0$ 可以替换为

$$x \in L^{(k)}, \quad y \in L^{(l)},$$

Proof. Suppose L is solvable. Then $\operatorname{ad}(L) \subseteq \mathfrak{gl}(L)$ is solvable. Then it follows from exercise 1.2.1 that $\operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) = 0$ for all $x \in L, y \in L'$.

Conversly, it follows from proposition 1.2.2 that $\operatorname{ad}(L')$ is solvable, then so dose L' and L . $\square$

1.3 Killing Form

Definition 1.3.1

Let L be a complex Lie algebra. The Killing form on L is the symmetric bilinear form defined by

$$\kappa(x, y) := \text{tr}(\text{ad } x \circ \text{ad } y), \quad \text{for } x, y \in L$$

Proposition 1.3.2

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

Theorem 1.3.3 ► Cartan's First Criterion

The complex Lie algebra L is solvable if and only if $\kappa(x, y) = 0$ for all $x \in L$ and $y \in L$.

Proof. Just a restatement of theorem 1.2.3 . □

Lemma 1.3.4

Suppose that L is a Lie algebra and I is an ideal of L . We write κ for the Killing form on L and κ_I for the Killing form on I , considered as a Lie algebra in its own right. If $x, y \in I$, then $\kappa_I(x, y) = \kappa(x, y)$.

Proof. We expand a basis of I to a basis of L . Since

$$\text{ad } x(L) \subseteq [x, L] \subseteq I$$

Then $\text{ad } x$ is represented of the form

$$\begin{pmatrix} A_x & B_x \\ O & O \end{pmatrix}$$

$\text{ad } y$ is represented of the form

$$\begin{pmatrix} A_y & B_y \\ O & O \end{pmatrix}$$

We have

$$\begin{aligned}
 \kappa(x, y) &= \text{tr} \left(\begin{pmatrix} A_x & B_x \\ O & O \end{pmatrix} \begin{pmatrix} A_y & B_y \\ O & O \end{pmatrix} \right) \\
 &= \text{tr} \left(\begin{pmatrix} A_x A_y & A_x B_y \\ O & O \end{pmatrix} \right) \\
 &= \text{tr}(A_x A_y) \\
 &= \kappa_I(x, y)
 \end{aligned}$$

□

1.3.1 Testing for Semisimplicity

Definition 1.3.5 ► perpendicular space

- Let β be a symmetric bilinear form on a finite-dimensional complex vector space V . If S is a subset of V , we define the **perpendicular space** to S by

$$S^\perp = \{x \in V : \beta(x, s) = 0 \text{ for all } s \in S\}.$$

- We say that β is **non-degenerate** if $V^\perp = 0$; that is, there is no non-zero vector $v \in V$ such that $\beta(v, x) = 0$ for all $x \in V$.

Proposition 1.3.6

If β is non-degenerate and W is a vector subspace of V , then

$$\dim W + \dim W^\perp = \dim V.$$

Proof. Define a linear map $\varphi : V \rightarrow V^*$,

$$\varphi(v) \mapsto (u \mapsto \beta(u, v))$$

Then

$$\ker \varphi = \{v \in V : \beta(u, v) = 0, \forall u \in V\} = V^\perp = 0$$

Since V is finite-dimensional, we have $\dim V = \dim V^*$. Then φ is an isomorphism. Define $W^0 \subseteq V^*$:

$$W^0 = \{\lambda \in V^* : \lambda(w) = 0, \forall w \in W\}$$

We have

$$\varphi(W^\perp) = W^0, \quad \dim W^\perp = \dim W^0$$

Let $i : W \hookrightarrow V$ the inclusion map, for each $\lambda^* \in V^*$, we define a map $i^* : V^* \rightarrow W^*$

$$i^*(\lambda)(w) = \lambda(i(w))$$

Then $\text{Im } i^* = W^*$, $\ker i^* = W^0$, we have

$$\dim W^0 + \dim W^* = \dim V^*$$

Then it follows that

$$\dim W^\top + \dim W^* = \dim V^* \implies \dim W^\perp + \dim W = \dim V$$

Since W, V have finite dimension. □

Exercise 1.3.7

Suppose I is an ideal of L . Shows that $I^\perp$ is an ideal of L .

Proof.

$$I^\perp = \{x \in L : \kappa(i, x) = 0, \forall i \in I\}$$

Take $x \in I^\perp, y \in L$, note that

$$[y, x] \in I^\perp \iff \kappa([x, y], i) = 0, \quad \forall i \in I$$

Since we have

$$\kappa(x, [y, i]) = 0, \forall y \in L, i \in I$$

Then

$$\kappa([x, y], i) = \text{tr}([\text{ad}_x, \text{ad}_y] \circ \text{ad}_i) = \text{tr}(\text{ad}_x \circ [\text{ad}_y \circ \text{ad}_i]) = \kappa(x, [y, i]) = 0$$

For all $i \in I$. That is $[x, y] \in I^\perp$. $[L, I^\perp] \subseteq I^\perp$. □

Corollary 1.3.8

If L is semisimple, then κ is non-degenerate.

Proof. Other wise, $L^\perp$ is an nonzero ideal of L . Since $(L^\perp)' \subseteq L$, we have

$$\kappa(L^\perp, (L^\perp)') \subseteq \kappa(L^\perp, L) = 0$$

Then it follows from the Cartan First Criteria that $L^\top$ is solvable. Then $L^\perp$ is a nonzero solvable ideal of L , which is a contradiction to the Semisimplicity of L . $\square$

Theorem 1.3.9 ► Cartan's Second Criterion

The complex Lie algebra L is semisimple if and only if the Killing form κ of L is non-degenerate.

1.4 Derivations of Semisimple Lie Algebras

Proposition 1.4.1

If L is a finite-dimensional complex semisimple Lie algebra, then $\text{ad } L = \text{Der } L$.

Proof. For each $x \in L$,

$$\begin{aligned} \text{ad } x [y, z] &= [x, [y, z]] \\ &= -[z, [x, y]] - [y, [z, x]] \\ &= [\text{ad } x (y), z] + [y, \text{ad } x (z)] \end{aligned}$$

$\text{ad } x$ is a derivation. ad is a Lie algebra homomorphism from L to $\text{Der } L$. For each $\delta \in \text{Der } L$,

$$\begin{aligned} [\delta, \text{ad } x]y &= \delta [x, y] - \text{ad } x (\delta y) \\ &= [\delta x, y] + [x, \delta y] - [x, \delta y] \\ &= \text{ad } (\delta x) (y) \end{aligned}$$

Thus $[\delta, \text{ad } x] = \text{ad } (\delta x)$, the image of ad is an ideal of $\text{Der } L$.

Now we bring in our assumption that L is complex and semisimple. First, note that $\text{ad} : L \rightarrow \text{Der } L$ is one-to-one, as $\ker \text{ad} = Z(L) = 0$, so the Lie algebra $M := \text{ad } L$ is isomorphic to L , therefore it is semisimple as well. We have

$$\dim M + \dim M^\perp = \dim \text{Der } L$$

so our aim is to show that $\dim M^\perp = 0$. As M is an ideal of $\text{Der } L$, the Killing form κ_M of M is the restriction of the Killing form on $\text{Der } L$. By Cartan's Second Criterion, κ_M is non-degenerate, so $M^\perp \cap M = 0$, and hence $[M^\top, M] = 0$. Thus if $\delta \in M^\perp$, and $\text{ad } x \in M$, then $[\delta, \text{ad } x] = 0$ for $M, M^\perp$ are ideals. But

we saw above that

$$[\delta, \text{ad } x] = \text{ad } (\delta x)$$

so, for all $x \in L$, we have $\delta(x) = 0$; in other words, $\delta = 0$. □

1.5 Abstract Jordan Decomposition

Proposition 1.5.1

Let L be a complex Lie algebra. Suppose that δ is a derivation of L with Jordan decomposition $\delta = \sigma + \nu$, where σ is diagonalisable and ν is nilpotent. Then σ and ν are also derivations of L .

Proof. For $\lambda \in \mathbb{C}$, let

$$L_\lambda = \{x \in L : (\delta - \lambda) x = 0\}$$

L decomposes as a direct sum of generalised eigenspaces

$$L = \bigoplus_\lambda L_\lambda$$

As σ acts diagonalisably, then λ -eigenspace of σ is L_λ . Take $x \in L_\lambda$, $y \in L_\mu$, then

$$\begin{aligned} (\delta - (\lambda + \mu) 1_L) [x, y] &= [\delta x, y] + [x, \delta y] - [\lambda x, y] - [x, \mu y] \\ &= [(\delta - \lambda 1_L) x, y] + [x, (\delta - \mu 1_L) y] \end{aligned}$$

$(\delta - \lambda 1_L)^{m+n} [x, y]$ is the linear combination of the form $[(\delta - \lambda 1_L)^i x, (\delta - \mu 1_L)^j y]$, which is zero. Thus $[x, y] \in L_{\lambda+\mu}$, so

$$\sigma([x, y]) = (\lambda + \mu) [x, y]$$

which is the same as

$$[\sigma(x), y] + [x, \sigma(y)] = [\lambda x, y] + [x, \mu y]$$

Thus σ is a derivation, and so is $\delta = \sigma + \nu$. □

Theorem 1.5.2

Let L be a complex semisimple Lie algebra. Each $x \in L$ can be written uniquely as $x = d + n$, where $d, n \in L$ are such that $\text{ad } d$ is diagonalisable, $\text{ad } n$ is nilpotent, and $[d, n] = 0$. Furthermore, if $y \in L$ commutes

with x , then $[d, y] = 0$ and $[n, y] = 0$.

Proof. Let $\text{ad } x = \sigma + \nu$, where $\sigma \in \mathfrak{gl}(L)$ is diagonalisable, $\nu \in \mathfrak{gl}(L)$ is nilpotent, and $[\sigma, \nu] = 0$. We know that σ and ν are derivations of the semisimple Lie algebra L . And we know that $\text{ad } L = \text{Der } L$, so there exists $d, n \in L$ such that $\text{ad } d = \sigma$ and $\text{ad } n = \nu$. As ad is injective and

$$\text{ad } x = \sigma + \nu = \text{ad } (d + n)$$

we get that $x = d + n$. Moreover, $\text{ad } [d, n] = [\text{ad } d, \text{ad } n] = 0$. The uniqueness of d and n follows from the uniqueness of the Jordan decomposition of $\text{ad } x$. Suppose that $y \in L$ and that $(\text{ad } x)y = 0$. Since σ and ν may be expressed as polynomials in $\text{ad } x$. Let

$$\nu = c_0 1_L + c_1 \text{ad } x + \cdots + c_r (\text{ad } x)^r$$

Applying ν to y we see that $\nu(y) = c_0 y$. But ν is nilpotent and $\nu(x) = c_0 x$, so $c_0 = 0$. Thus $\nu(y) = 0$, and so $\sigma(y) = (\text{ad } x - \nu)y = 0$ also. $\square$

Definition 1.5.3

Let L be a semisimple Lie algebra and let $\theta : L \rightarrow \mathfrak{gl}(V)$ be a representation of L . Suppose that $x \in L$ has abstract Jordan decomposition $x = d + n$. Then the Jordan decomposition of $\theta(x) \in \mathfrak{gl}(V)$ is $\theta(x) = \theta(d) + \theta(n)$.